

Imran Khan, PhD

Digital Innovation & Technology Program Manager at Schneider Electric

I make enterprise data trusted and easy to use, and I connect data architecture to business decisions.

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6 months → 4 hours Product data processing time after the knowledge graph platform I designed	18 years In data, IoT and cloud architecture, 11 at Schneider Electric	7 patents 3 granted in the US, 4 pending	Senior Member IEEE (Institute of Electrical and Electronics Engineers), since 2021
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PROFILE

I work where data architecture meets business strategy. For 18 years, 11 of them at Schneider Electric, I have designed how data is modeled, governed and delivered, and I have helped leaders agree on what that data should do for the business.

Today I help Schneider's IT and IoT (Internet of Things) data domains raise their data quality and meet the company's data rules. I also lead the case for knowledge graphs across the enterprise, work that is presented to the Executive Committee.

Before that, I designed the data model behind a new electrical product data platform. It cut data processing time from six months to four hours. I have professional experience of working in France and Canada and have led technical teams across the USA, Europe and India. I am at my best in ambiguous situations where technology, people and strategy have to line up, and I explain technical choices in plain terms so decision-makers can act on them.

EXPERIENCE

Schneider Electric,
Grenoble, France.
Last four roles,
newest first.

Digital Innovation & Technology Program Manager

Jan 2025 to present

Data & Performance team, Group Governance

- Help the IT and IoT data domains raise their data maturity and meet the company's golden data rules.
- IT domain result: better tracking of assets and their financial cost, and data that internal processes can use directly.
- IoT domain result: data from digital offers that is traceable and meets the regulations Schneider must follow.
- Lead the enterprise strategy for formal semantics and knowledge graphs, presented to the Executive Committee.
- Use AI to run golden data rule compliance checks on critical IT systems.
- Help set up the global business data catalog so teams across the company can find, trust and reuse data.
- Work with senior leaders, enterprise architects, offer development teams and data domain owners to tie data work to business outcomes.

Technical Data Domain Expert and Architect

Mar 2024 to Dec 2024

Interoperability platform for electrical design software

- Led the move from a legacy data model to a semantic model (an ontology) so that several electrical design tools can exchange project data.
- Worked with electrical domain experts to capture what each project must contain, and set the data quality practices.
- Defined, with engineering, the services and APIs (Application Programming Interfaces) that keep project data consistent when many people edit it with different tools.

- Designed the semantic model (using the W3C standards SKOS, RDF and OWL) for a next-generation product information platform used by electrical design software.
- Result: electrical product data processing time dropped from six months to four hours.
- Owned the technical architecture for storing and processing the data.
- Managed budget, hiring and a team split between India and France.
- Advised on the group data strategy, including AVEVA.

- Led the Knowledge Graph Program and a team of five, delivering solutions for the Industrial Automation and Energy Management business units.
- Use cases in oil and gas, process automation, industrial IoT and machine solutions.
- Built an internal community that taught teams what knowledge graphs can do and fed new use cases into delivery.

EXPERTISE

From data rules to software products, and the people who need to agree on both.

Data Governance and Management

Data standards, data quality, golden data rules, business data catalogs, data domain operating models.

Software Product Design and Development

Led technical teams that designed, built and shipped software products, from requirements to release. Owned technical design and architecture; hands-on in Java, C and Python, and in cloud services on Microsoft Azure.

Leadership and Strategy

Executive stakeholder alignment, change management, program and project management, budget and hiring, teams spread across the USA, Europe and India.

Tools and methods: Git · Jira · Azure DevOps · Agile and SAFe (Scaled Agile Framework)

Data and Knowledge Architecture

End-to-end dataflow design, semantic data models, ontologies, knowledge graphs (RDF, SKOS, OWL), interoperability between systems.

AI and Large Language Models (LLMs)

Preparing enterprise data so AI can use it: quality, meaning and governance, with knowledge graphs as a trusted source of facts for LLMs. Daily, hands-on use of LLM assistants and AI agents for analysis, drafting and task automation. Example: using AI to run golden data rule compliance checks on critical IT systems.

Domain Knowledge

Energy management and industrial automation: buildings, data centers, process automation, low-voltage electrical products and electrical network design.

PATENTS

7 patents: 3 granted, 4 pending

Dynamic ontology data operation	US 12,111,861 B2 · 2024
Application programming interface enablement of consistent ontology model instantiation	US 11,972,357 B2 · 2024
Semantic search method for a distributed data system with numerical time series data	US 11,526,510 B2 · 2022

Pending: 4 applications, including Plant infrastructure modelling (US 2023/0401347 A1, published Dec 2023).

EDUCATION

PhD in Computer Science and Networks

Télécom SudParis and Université Pierre et Marie Curie (Paris VI), 2012 to 2015

Thesis: cloud-based, cost-efficient application and service provisioning in virtualized wireless sensor networks.

LANGUAGES

English · full professional

French · working proficiency

Urdu · native

RESEARCH

Even after joining industry, I continued my research in IoT, cloud computing and semantic data. My papers have been cited more than 2,900 times. [See publications on Google Scholar](#)